Flame Tests

What is the flame test?

The flame test is used to visually determine the identity of an unknown metal of an ionic salt based on the characteristic colour the salt turns the flame of a Bunsen burner.

What are the limitations of this test?

The value of the flame test is limited by interference from other brighter colours and by ambiguities where certain different metals cause the same flame colour.

What is happening?

When the salt is placed into the flame, the electrons from the metal become excited and jump up energy levels. When they fall back to their ground state they emit the energy they absorbed as photons of light. Often these photons are in the visible spectrum and a colour unique to that ion can be seen in the flame.

Method

1. Light your Bunsen burners so that a hot flame is produced
2. Collect a pop stick and dip it in distilled water. Place it in the blue flame, and then remove it. Observe the colour of the flame and record it in your table.
3. Next using a fresh popstick, dip the stick into the distilled water and then into the Barium chloride. Record the colour in your table.
4. Repeat for all of the metal chlorides, **using a fresh popstick each time**.

Results

Salt	Metal Ion	Non Metal ion	Flame Colour
Distilled Water			
Barium Chloride	Ba^{2+}	Cl^{1-}	
Nickel Chloride	Ni^{2+}	Cl^{1-}	
Potassium Chloride	K^{+}	Cl^{1-}	
Copper Chloride	Cu^{2+}	Cl^{1-}	
Strontium Chloride	Sr^{2+}	Cl^{1-}	
Sodium Chloride	Na^{+}	Cl^{1-}	

Answer the questions below:

Questions

1. What was the purpose of using the stick just soaked in distilled water?

2. Why did the water have to be distilled and not from the tap?

3. Why do you think different metal ions produce different flame test colours?

4. How is this test limited?

5. Describe how flame testing could be used as a method of identifying unknown metal salts?

6. Where did the electrons get the energy to jump levels?

7. Which of the solutions would you use to make a firework:

    a) Red?
    b) Green?
    c) Blue/green?

8. This test is described as a qualitative test. What do you think this means?

9. Most chemistry experiments are called quantitative experiments. What do you think this means?